

Sample Output — Per-Agent I/O

Two cases traced end-to-end: one auto-cleared, one escalated — with final reports

Document	Sample Output (Agent I/O & Final Reports)	Version	1.2
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This document traces two audit cases through every agent, showing the concrete input each agent receives and the structured output it returns, ending with the final case report and audit entry. **Case A** auto-clears; **Case B** escalates to a partner after the Verifier catches a hallucination.

Pipeline at a glance

Agent	Consumes	Produces
Intake pre-filter	GL CSV row	Transaction + flags[] (or dropped)
Supervisor · Planner	case	plan · routing · escalation
Evidence agent	case	evidence{} + citations
Challenger	case + evidence	challenger_view (concern)
Defender	case + evidence	defender_view (legitimacy)
Adjudicator	evidence + both views	adjudication{risk·conf·disposition}
Verifier · QA	adjudication + evidence	verification{passed·issues}
Confidence gate	adjudication + verification	routing{auto-review·manual}
Human review	full bundle + transcript	review{action·comment·signature}
Report / Audit	everything	case file + audit_history[]

Case A — Auto-cleared

A legitimate recurring rent that trips only the soft `fx_conversion` flag. Fully grounded; the confidence gate auto-clears it with **no human review**.

The case (input to the pipeline)

```
case = {"case_id":"CASE-0042","txn_id":"TXN-0003","date":"2026-06-25","account":"Rent",
"vendor":"Brookfield Properties","description":"Monthly office rent – Mumbai",
"currency":"INR","amount":1558481.67,"fx_rate_used":0.012,"amount_usd":18701.78,
"posted_by":"jsmith","flags":["fx_conversion"]}
```

1 · Supervisor plans & routes

INPUT

```
case  # flagged transaction above
```

OUTPUT

```
plan = {"steps":["gather_evidence","run_debate","adjudicate","verify"],
        "specialists":["evidence","challenger","defender","adjudicator","verifier"],
        "max_debate_rounds":2,"materiality_usd":25000,
        "escalation":"none (amount < materiality)",
        "route_hint":"soft FX flag -> expect auto-clear if grounded"}
```

2 · Evidence agent — RAG + history + FX

INPUT

```
case
```

OUTPUT

```
evidence = {
  "policy":[{"id":"LSE-01 §3","v":"2.0","text":"Recorded rent must match the signed lease
schedule."},
    {"id":"FX-04 §1","v":"1.2","text":"Foreign-currency postings use the daily rate ±2%."}],
  "history":{"vendor_prior_12m":11,"account_mean_usd":18650,"z_score":0.04},
  "external":{"fx":{"INR->USD@2026-06-25":0.01201,"posted":0.012,"delta":"0.1% (within ±2%)"},
    "vendor_master":"Brookfield Properties – APPROVED (lease LSE-1188)"},
  "citations":["LSE-01 §3","FX-04 §1","history:Rent","fx:frankfurter","vendormaster"],
  "api_metadata":[{"source":"frankfurter","status":200,"latency_ms":210,"cached":false}]}
```

3 · Challenger (skeptical)

INPUT

```
case + evidence
```

OUTPUT

```
challenger_view = {"position":"concern","severity":"low",
  "claims":["Large foreign-currency posting (INR 1.56M) warrants a rate check.",
    "Amount is above the account mean, though only marginally (z=0.04)."],
  "red_flags":["fx_conversion"]}
```

4 · Defender (independent)

INPUT

```
case + evidence    # does NOT see the Challenger's reasoning
```

OUTPUT

```
defender_view = {"position":"legitimate_possible","confidence_in_legitimacy":0.95,
  "arguments":["Approved lessor (LSE-1188); 11 matching monthly rents in 12 months.",
    "FX rate 0.012 within 0.1% of the daily reference (FX-04 §1).",
    "amount_usd sits on the account mean (z=0.04) – not an outlier."],
  "conditions":["lease schedule on file"]}
```

5 · Adjudicator weighs both

INPUT

```
evidence + challenger_view + defender_view
```

OUTPUT

```
adjudication = {"risk_level":"Low","confidence":0.94,
  "disposition":"clear_recurring_legitimate",
  "rationale":"Approved lessor, recurring identical rent, FX within tolerance, on-mean amount; "
    "the only flag (fx_conversion) is fully explained by grounded evidence.",
  "citations":["LSE-01 §3","FX-04 §1","history:Rent z=0.04","fx:frankfurter:
    0.1%","vendormaster:approved"]}
```

6 · Verifier checks grounding

INPUT

```
adjudication + evidence
```

OUTPUT

```
verification = {"passed":true,"checked_claims":4,"grounded":4,"ungrounded":0,
  "issues":[], "action":"proceed"}
```

7 · Confidence gate routes

INPUT

```
adjudication + verification + amount
```

OUTPUT

```
routing = {"decision":"auto",
  "reason":"confidence 0.94 ≥ 0.90 AND verified AND risk=Low AND 18,701.78 < 25,000"}
# no human review — straight to report (QA-sampled)
```

Final report — Case A

Verdict	LOW CASE-0042 · TXN-0003 · Brookfield Properties · Rent · \$18,701.78
Date / posted by	2026-06-25 · jsmith · INR 1,558,481.67 @ 0.012 = USD 18,701.78
Flags	fx_conversion
Evidence	Approved lessor (LSE-1188) · 11 prior monthly rents · FX within 0.1% · on-mean (z=0.04)
Debate — concern	FX posting warrants a rate check (severity low)
Debate — defence	recurring contractual rent, rate within tolerance (conf 0.95)
Verdict	Low · confidence 0.94 · disposition: clear (recurring legitimate)
Verifier	passed — 4/4 claims grounded
Routing	auto-cleared — QA-sampled, no human review

```
audit_history += {"event":"case_closed","case_id":"CASE-0042",
  "verdict":"Low / clear_recurring_legitimate","confidence":0.94,"route":"auto_cleared",
```

```
"human":null,"sources_used":["LSE-01 §3","FX-04 §1","fx:frankfurter","vendor_master"],
"timestamp":"2026-06-26T09:14:02Z","hash_prev":"...","hash":"..."}
```

Case B — Escalated (Verifier catches a hallucination)

A \$58,000 related-party payment to “Mehta Family Holdings” (Consulting). The Adjudicator's first pass invents an approval; the Verifier catches it and forces a re-run; the verdict then escalates to a partner.

The case (input to the pipeline)

```
case = {"case_id":"CASE-0007","txn_id":"TXN-0876","date":"2026-05-20","account":"Consulting",
"vendor":"Mehta Family Holdings","description":"Consulting - related party (director J.
Mehta)",
"currency":"USD","amount":58000,"amount_usd":58000,"posted_by":"rkumar",
"flags":["materiality","round-dollar amount","statistical outlier (Consulting)",
"related-party indicator","unknown vendor"]}
```

1 · Supervisor

INPUT

```
case
```

OUTPUT

```
plan = {"steps":["gather_evidence","run_debate","adjudicate","verify"],
"specialists":["evidence","challenger","defender","adjudicator","verifier"],
"max_debate_rounds":2,"materiality_usd":25000,
"escalation":"partner_if_material","route_hint":"material -> expect manual / partner"}
```

2 · Evidence agent

INPUT

```
case
```

OUTPUT

```
evidence = {
  "policy":[{"id":"RP-01 §2","v":"2.1","text":"Related-party txns require independent review (ISA
550)."},
    {"id":"AP-02 §1","v":"3.0","text":">>$5k needs manager approval; >$25k needs partner
approval."}],
  "history":{"vendor_prior_12m":0,"account_mean_usd":15126,"z_score":4.1},
  "external":{"fx":"n/a (USD)","vendor_master":"Mehta Family Holdings – NOT in approved vendor
master"},
  "citations":["RP-01 §2","AP-02 §1","history:Consulting","vendormaster"],
  "api_metadata":[{"source":"vendor_master","status":200,"cached":false}]}
```

3 · Challenger

INPUT

```
case + evidence
```

OUTPUT

```
challenger_view = {"position":"concern","severity":"high",
  "claims":["Vendor implies related party to director J. Mehta (RP-01 §2).",
    "$58,000 is 4.1σ above the Consulting mean AND an exact round number.",
    "Vendor absent from approved master; no 12-month history (ghost vendor).",
    "$58,000 exceeds the $25,000 partner-approval threshold (AP-02 §1)."],
  "red_flags":["related_party","round_dollar","outlier","ghost_vendor","approval_breach"]}
```

4 · Defender (independent)

INPUT

```
case + evidence    # does NOT see the Challenger's reasoning
```

OUTPUT

```
defender_view = {"position":"legitimate_possible","confidence_in_legitimacy":0.25,
  "arguments":["Large one-off strategy retainers occur in Consulting; can be round by contract.",
    "Related-party is permitted IF independently approved and disclosed.",
    "A new vendor naturally has no 12-month history."],
  "conditions":["signed contract","independent approval","related-party disclosure on file"]}
```

5 · Adjudicator (round 1)

INPUT

```
evidence + challenger_view + defender_view
```

OUTPUT (note the unsupported sign-off claim)

```
adjudication = {"risk_level":"High","confidence":0.86,"disposition":"investigate_before_reliance",
  "rationale":"... outweighs the benign-retainer hypothesis. Director sign-off obtained 2026-05-19.",
  "citations":["RP-01 §2","AP-02 §1","history:Consulting z=4.1","vendormaster:absent"]}
```

6 · Verifier (round 1) — FAIL

INPUT

```
adjudication + evidence
```

OUTPUT

```
verification = {"passed":false,"checked_claims":5,"grounded":4,"ungrounded":1,
  "issues":["'Director sign-off obtained 2026-05-19' has NO supporting evidence id (hallucination)."],
  "action":"revise"}    # loop back to adjudicator (round < 2)
```

5' · Adjudicator (round 2, revised)

OUTPUT

```
adjudication = {"risk_level":"High","confidence":0.86,"disposition":"investigate_before_reliance",
  "rationale":"Related-party + 4.1σ round-number outlier + ghost vendor + breach of AP-02 "
```

```
"partner-approval threshold; NO approval or disclosure evidence on file.",
" Citations":["RP-01 §2","AP-02 §1","history:Consulting z=4.1","vendormaster:absent"]}]}
```

6 · Verifier (round 2) — PASS

OUTPUT

```
verification = {"passed":true,"checked_claims":4,"grounded":4,"ungrounded":0,"issues":
[],"action":"proceed"}
```

7 · Confidence gate

OUTPUT

```
routing = {"decision":"manual",
"reason":"amount $58,000 ≥ materiality $25,000 AND risk=High -> partner escalation"}
```

8 · Human review — durable interrupt() / resume()

INPUT

```
full bundle: case + evidence + debate_transcript + adjudication + verification
```

OUTPUT

```
review = {"reviewer_id":"a.sharma (Partner)","action":"reject",
"comment":"Hold payment. Obtain signed contract + independent approval + related-party "
"disclosure before any reliance.",
"signature":"sha256:9f2c4b...e71","timestamp":"2026-06-25T11:42:08Z"}
```

Final report — Case B

Verdict	HIGH CASE-0007 · TXN-0876 · Mehta Family Holdings · Consulting · \$58,000
Date / posted by	2026-05-20 · rkumar · USD 58,000.00
Flags	materiality · round-dollar · outlier (z=4.1) · related-party · unknown vendor
Evidence	Vendor NOT in approved master · 0 prior 12-mo activity · account mean \$15,126
Debate — concern	related party (RP-01 §2) · 4.1σ round outlier · ghost vendor · breach of AP-02 \$25k partner approval
Debate — defence	possible one-off retainer, but no contract / approval / disclosure on file (conf 0.25)
Verifier	round 1 FAIL (caught hallucinated sign-off) → re-run → round 2 passed 4/4
Verdict	High · confidence 0.86 · disposition: investigate before reliance
Routing	manual — partner escalation (≥ materiality \$25,000)
Decision	a.sharma (Partner): reject — hold payment; obtain contract + approval + disclosure

```
audit_history += {"event":"case_closed","case_id":"CASE-0007",
"verdict":"High / investigate_before_reliance","confidence":0.86,"rounds":2,
"route":"manual","escalated_to":"partner",
"debate":{"challenger":
```

```
[ "related_party", "round_dollar", "outlier", "ghost_vendor", "approval_breach",  
  "defender": { "position": "legitimate_possible", "confidence": 0.25 } },  
  "verification": { "round1": "fail (1 ungrounded)", "round2": "pass 4/4" },  
  "human": { "by": "a.sharma(Partner)", "action": "reject", "signed": true, "ts": "2026-06-25T11:42:08Z" },  
  "sources_used": [ "RP-01 §2", "AP-02 §1", "vendor_master", "history:Consulting" ],  
  "timestamp": "2026-06-25T11:42:09Z", "hash_prev": "...", "hash": "..." }
```

Why both cases matter for the demo. Case A proves the **auto-clear** path and that a soft flag with grounded evidence does not waste a reviewer's time. Case B proves the **Verifier catches a hallucination**, the **bounded re-run** loop, and **mandatory partner escalation** on a material high-risk verdict — the headline behaviours of the multi-agent design.